

Backend (brain/ package) — a real forecast → fuse → decide pipeline, not a staged script:

- traffic_source.py — deterministic synthetic ground-truth speeds per road edge, standing in for a live feed until macro/perception are wired in. A "congestion scenario" toggle simulates a live disruption that a pre-trained historical model can't see but a recent-data model can.

- forecasting.py — wraps the two existing forecasters (HistoricalAverageForecaster, SpatialTemporalForecaster), converts each into the shared RoadObservation contract, and fuses them. Any forecaster can fail or be disabled and the estimate just narrows to whatever's left.

- engine.py (DecisionEngine) — sums fused per-edge travel times into route ETAs, applies the existing hysteresis (should_switch_route) to pick a route, and drives three IntersectionController state machines along the selected route to produce real signal phases — not hardcoded ones.

- simulation.py (BrainSimulation) — stateful wrapper the API ticks, same shape as sim/demo.py's DemoSimulation.

Generalized fusion (fusion/engine.py) — now takes any number of observations (still backward-compatible with existing 2-arg calls), so it's ready for perception/macro to join later without changing the brain's logic.

API — GET /api/brain/snapshot, PATCH /api/brain/controls, POST /api/brain/reset, WS /ws/brain.

Web — new "Brain" tab (BrainPage.tsx) reusing the existing map/styling, with live toggles for each forecaster and the congestion scenario, so you can watch the route/signal decisions change in real time as models go on and off.

Verified: both models → reroutes on congestion; spatial-temporal alone → reroutes fast; historical alone → correctly lags (it can't see live disruptions, which is realistic); zero models → still produces a decision via static fallback. Full existing test suite still passes.

Two caveats: I couldn't run npm install/npm run build or boot the FastAPI server here (no node_modules, no fastapi installed in this Python), so the frontend and wiring are

unverified by an actual browser/server run — want me to install deps and do that pass, or will you run it locally?